

A global condition that is really a local one: OEIS A223056

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Abstract

OEIS A223056 carries an empirical recurrence of order 4 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry asks that every connected set of equal values lie in a straight line, which is a condition on components and so on the array as a whole. It is shown here to be equivalent to a condition on single pairs of adjacent rows: the array is admissible exactly when no cell has an equal neighbour beside it and an equal neighbour above or below it, and that in turn is decided one row at a time. The counted arrays are then the walks in a finite digraph on $S = 5$ vertices, $a(n)$ is C -finite, and whether the conjectured recurrence annihilates it is a finite exact computation with Cayley–Hamilton bounding the work.

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1 The conjecture

OEIS A223056 is “Number of $n \times 2$ 0..3 arrays with all horizontally or vertically connected equal values in a straight line, and new values 0..3 introduced in row major order.”. It has offset 1 and begins

2, 10, 114, 1473, 19360, 255045, 3361024, 44294253, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 18a(n-1) - 71a(n-2) + 102a(n-3) - 48a(n-4)$ for $n > 5$.

As of the “Last modified” line on the live entry (Aug 16 2018 16:06:49, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The condition is not global

The entry counts arrays with 2 columns over the alphabet $\{0, 1, 2, 3\}$, and asks that every set of equal values joined horizontally or vertically lie in a straight line. As stated this is a condition on connected components, and a component can run the length of the array; nothing about it is confined to a few rows. The first lemma replaces it by a condition on single cells, and the second by a condition on pairs of consecutive rows.

Call two cells *joined* when they are horizontally or vertically adjacent and hold the same value, and let a *component* be a connected set of cells under that relation. A *turn* is a cell with a joined neighbour to its left or right *and* a joined neighbour above or below.

Lemma 1. *Every component lies in a single row or a single column if and only if the array has no turn.*

Proof. A turn lies in a component containing a cell to its side and a cell above or below it, so that component meets two rows and two columns and lies in neither a single row nor a single column.

Conversely, suppose some component C lies in no single row and no single column. C is connected, so it has at least two cells and hence at least one joining edge; if every edge of C were horizontal then all of C would lie in one row, and if every edge were vertical it would lie in one column. So C has a horizontal edge e and a vertical edge f . Take a path of edges of C from e to f ; along it the type changes somewhere, so two consecutive edges of the path have different types. Consecutive edges of a path share a cell, and that shared cell has a joined neighbour beside it and one above or below it. It is a turn. $\square$

The point of Lemma 1 is that a turn is visible in one cell and its four neighbours. That already bounds the condition, but it bounds it to three rows. It can be brought down to two.

For a row r write $B(r) \subseteq \{1, \dots, 2\}$ for the set of positions j with $r_{j-1} = r_j$ or $r_j = r_{j+1}$ — the cells of that row with a joined neighbour beside them. $B(r)$ depends on r alone.

Lemma 2. *An array has no turn if and only if every pair of consecutive rows r, s satisfies*

$$r_j \neq s_j \quad \text{for every } j \in B(r) \cup B(s).$$

Proof. A cell in row i at position j is a turn exactly when $j \in B(r^{(i)})$ and it equals the cell directly above or directly below it. If some pair of consecutive rows $r = r^{(i)}$, $s = r^{(i+1)}$ has $r_j = s_j$ with $j \in B(r)$, that cell of row i is a turn; with $j \in B(s)$, that cell of row $i+1$ is a turn. Conversely a turn at row i , position j , has $j \in B(r^{(i)})$ and agrees with row $i-1$ or row $i+1$ at j , and the corresponding consecutive pair violates the displayed condition. $\square$

So the array is admissible precisely when consecutive rows are compatible in the sense of Lemma 2, and compatibility is decided by the two rows and nothing else. The state is one row. There is no window to carry, no judgement deferred to a later row, and no special treatment of the last row: a one-row array has no vertical neighbours at all and is always admissible, which is where these sequences start.

The entry also asks that new values be introduced in row major order. That is not a condition on any pair of rows — whether a value may appear here depends on the whole array so far — but the whole prefix enters only through one integer, how many values have been introduced already, so the state is a row together with a number in $\{0, \dots, 4\}$. The effect is that each array is counted once for the whole class of arrays differing from it only by renaming values, which is what the entry counts.

Make those states the vertices of a digraph, with an edge from r to s when the pair is compatible and the counter advances legally. Merging vertices with identical futures leaves $S = 5$. An admissible array is exactly a walk, so with M the adjacency matrix and ι, τ the vectors recording which states may begin and end an array,

$$a(n) = \iota^\top M^{n-1} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 3. *Let $q(t) = t^4 - \sum_i c_i t^{4-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+4} - \sum_i c_i M^{j+4-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. A constant factor in front of a divides out of the residual and changes nothing here. $\square$

Theorem 4.

$$a(n) = 18a(n-1) - 71a(n-2) + 102a(n-3) - 48a(n-4)$$

for every $n > 5$.

Proof. The vector $w = q(M)\tau$ was formed by 4 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 5 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The bound is exact: at $n = 5$ the recurrence fails on the entry’s own published terms, so no larger range holds. $\square$

4 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry’s English, and a misreading gives different counts.

Second, the reduction of Lemmas 1 and 2 was checked against the thing it replaces. A separate program enumerates every array of the given shape one by one, finds the connected sets of equal values by an ordinary graph search, and tests whether each lies in a single row or a single column — the entry’s condition as written, with no reduction at all. Its counts agree with the model’s. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 3 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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